Density-Aware Adaptive Traffic Light System (using CV and priority based scheduling) – Anoushka Basak (24BAI0153)

Abstract:

Urban traffic congestion is worsened by fixed-time traffic signals that fail to account for real-time variation in vehicle density across intersections. This project proposes a traffic monitoring and adaptive signal control system that uses classical computer vision techniques to estimate vehicle density at each lane of an intersection and dynamically allocate green-light duration using a weighted priority scheduling algorithm, without relying on machine learning models.

Dataset: The system will use recorded/simulated traffic footage (e.g., open-source intersection surveillance datasets such as UA-DETRAC, or self-recorded/simulated datasets from platforms like SUMO for controlled testing). Each video feed represents one lane approaching an intersection, providing frame-by-frame visual input for vehicle detection.

Application: The core application is a real-time signal control simulator: given live or recorded footage from single/multiple lanes, the system estimates congestion per lane and computes signal timing dynamically, aiming to reduce average wait time compared to fixed-cycle signals. This could be positioned as a low-cost retrofit for existing traffic infrastructure using standard CCTV cameras.

Methodology:

1. **Preprocessing** — background subtraction (e.g., MOG2) or frame differencing to isolate moving/static vehicles from each lane's video feed.
2. **Vehicle detection/counting** — contour detection and blob analysis (using OpenCV's findContours, morphological operations) to count vehicles and estimate lane occupancy.
3. **Density estimation** — converting vehicle count and occupied pixel area into a normalized congestion score per lane.
4. **Weighted priority scheduling** — a scheduling algorithm (similar to weighted round-robin) that assigns green-light duration proportional to each lane's congestion score, with a minimum guaranteed time per lane to prevent starvation.
5. **Evaluation** — comparing average wait time and throughput of the adaptive system against a fixed-time baseline using simulated traffic scenarios.